fsoft2026 1DD: Iteration 1 – Technical Report Domain Model, Glossary, Functional and Non-Functional Requirements, System Sequence Diagram

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Abstract. In the following pages, the reader will be able to be informed about the application's domain model, glossary, functional and non-functional requirements, containing use cases and their specifications, and system sequence diagrams (SSD) for the MedManager project.

Keywords: Business, C++, Vaccination Center, MedManager, Management, Software Development, Command-line Interface, Health

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1 Introduction

The MedManager project is a stand-alone management system designed for a single local vaccination center. It focuses on the control of physical vaccine stock, appointment scheduling, and the end-to-end vaccination process.

We named the application being developed in this project "**MedManager**". The system's architecture incorporates four key stakeholder interfaces, each serving distinct but interconnected roles within the MedManager ecosystem:

- **Center Administrator:** Responsible for internal management, including specifying vaccine types, registering physical doses, and managing employee records.
- **Receptionist:** Responsible for registering SNS users, scheduling administrations, and registering patient arrivals.
- **Nurse:** Responsible for consulting the waiting room and recording the administration of vaccines.
- **SNS User:** Intends to use the application to schedule a vaccine administration.

The objective of MedManager is to provide a reliable solution that optimizes all operations related to the vaccination process at a local center. The system will be developed in C++, which is the programming language taught in this class.

2 Domain Model

Fig. 1. MedManager's Domain Model

3 Glossary

The following glossary defines the core technical terms used throughout the MedManager system.

Table 1. MedManager Glossary

Term/Expression (EN)	Termo (PT)	Definition/Description
Center Administrator	Administrador do Centro	Employee responsible for managing and configuring the healthcare center's capacities and settings.
Appointment	Agendamento	A scheduled slot (date and time) for an SNS User to receive a specific type of vaccine.
Healthcare Center	Centro de Saúde	The physical facility where vaccinations take place, defined by its operating hours and hourly capacity.
Nurse	Enfermeiro	Healthcare professional responsible for administering vaccines and recording the vaccination process details.
Employee	Funcionário	General entity representing anyone working at the center, identified by ID, name, and phone number.
Vaccination Process	Processo de Vacinação	The actual record of vaccine administration, including lot number and adverse reactions.
Receptionist	Rececionista	Employee who manages user registration and confirms appointments upon the user's arrival.
Vaccine Type	Tipo de Vacina	A category of vaccines defined by the disease they target (e.g., COVID-19, Influenza).
SNS User	Utente do SNS	A citizen registered in the National Health Service who is eligible to receive vaccinations.
Vaccine	Vacina	The specific commercial product and brand used during the vaccination process.

4 Non-Functional Requirements

4.1 Usability

- Command-line based user interface
- Informative, yet simple, error messages
- Easy-to-use option menus

4.2 Reliability

- Data storage must be reliable
- System must be available at any time
- Validation of all user inputs, including Portuguese-format phone numbers and citizen card numbers

4.3 Performance

- Response time under 1 second for all operations

4.4 Supportability

- Full application support through GitHub repository
- Recurrent application testing

4.5 Design Constraints

- Data persistent and kept in binary files
- No duplicate entries for the same SNS user on the same day

4.6 Implementation Constraints

- C++ programming language

4.7 Interface Constraints

- No graphical interface; interaction is exclusively via command-line

5 Functional Requirements

Functional requirements are captured through Use Case Diagrams and their corresponding specifications.

5.1 Actor: Center Administrator

Fig. 2. Center Administrator's Use Case Diagram

Table 2. Use Case: Specify New Vaccine Type (US10)

Actor	Center Administrator
Use case name	Specify New Vaccine Type
Description	The administrator creates a new type of vaccine in the system.
Precondition	The "Center Administrator" user must be selected.
Post-condition	A new VaccineType is saved in the system.
Main flow	→ System displays an option menu → User selects "Specify New Vaccine Type" → System displays an input space → User fills the input space with the vaccine type name and target disease → System checks if the information type is correct → A new VaccineType is added to the system
Alternative path	– Information type is incorrect → User receives an error message → Use case resumes at step 3
Exceptions	If the user, at any point, wishes to cancel, the user can type "cancel operation" and the use case resumes at step 1

Use Case: Specify New Vaccine Type – US10 (Specification)

Fig. 3. SSD: Center Administrator's UC 1

SSD: UC 1 – Specify New Vaccine Type

Table 3. Use Case: Register Physical Vaccine Dose (US11)

Actor	Center Administrator
Use case name	Register Physical Vaccine Dose
Description	The administrator adds a physical vaccine dose to the center's inventory, including lot number, brand, and expiration date.
Precondition	The "Center Administrator" user must be selected and at least one VaccineType must exist in the system.
Post-condition	A new Vaccine dose is saved and linked to an existing VaccineType.
Main flow	→ System displays an option menu → User selects "Register Physical Vaccine Dose" → System displays the list of existing vaccine types → User selects a vaccine type → System displays an input space → User fills the input space with lot number, brand, and expiration date → System checks if the information type is correct → A new Vaccine dose is added to the inventory
Alternative path	– Information type is incorrect → User receives an error message → Use case resumes at step 5 – No vaccine types exist in the system → System sends a "No Vaccine Types" message → Use case resumes at step 1
Exceptions	If the user, at any point, wishes to cancel, the user can type "cancel operation" and the use case resumes at step 1

Use Case: Register Physical Vaccine Dose – US11 (Specification)

Fig. 4. SSD: Center Administrator's UC 2

SSD: UC 2 – Register Physical Vaccine Dose

Table 4. Use Case: List Vaccine Stock (US12)

Actor	Center Administrator
Use case name	List Vaccine Stock
Description	A list of all physical vaccine doses available in the center is shown to the administrator, grouped by type and sorted alphabetically by brand.
Precondition	The "Center Administrator" user must be selected and at least one vaccine dose must be registered in the system.
Post-condition	None
Main flow	→ System displays an option menu → User selects "List Vaccine Stock" → System checks if any vaccine doses are registered → System displays all doses grouped by VaccineType and sorted alphabetically by brand
Alternative path	– No vaccine doses are registered in the system → System sends a "No Vaccines" message → Use case resumes at step 1
Exceptions	None

Use Case: List Vaccine Stock – US12 (Specification)

Fig. 5. SSD: Center Administrator's UC 3

SSD: UC 3 – List Vaccine Stock

Table 5. Use Case: Register New Employee (US14)

Actor	Center Administrator
Use case name	Register New Employee
Description	The administrator registers a new employee in the system, assigning them a role (Nurse, Receptionist, or Administrator).
Precondition	The "Center Administrator" user must be selected.
Post-condition	A new Employee is saved in the system with a unique phone number and e-mail address.
Main flow	→ System displays an option menu → User selects "Register New Employee" → System displays an input space → User fills the input space with name, phone number, e-mail address, citizen card number, and role → System checks if the information type is correct and if phone number and e-mail are unique → Use case returns to step 3 until all mandatory properties are filled → A new Employee is added to the system
Alternative path	– Information type is incorrect or phone/CC not in Portuguese format → User receives an error message → Use case resumes at step 3 – Phone number or e-mail already exists in the system → User receives a "Duplicate Data" error message → Use case resumes at step 3
Exceptions	If the user, at any point, wishes to cancel, the user can type "cancel operation" and the use case resumes at step 1

Use Case: Register New Employee – US14 (Specification)

Fig. 6. SSD: Center Administrator's UC 4

SSD: UC 4 – Register New Employee

Table 6. Use Case: List Employees by Role (US15)

Actor	Center Administrator
Use case name	List Employees by Role
Description	A list of all employees assigned to a specific role is shown to the administrator.
Precondition	The "Center Administrator" user must be selected and at least one employee must be registered in the system.
Post-condition	None
Main flow	→ System displays an option menu → User selects "List Employees by Role" → System displays the available roles → User selects a role → System checks if any employees are assigned to that role → System displays a list of all employees with the selected role
Alternative path	– No employees are assigned to the selected role → System sends a "No Employees" message → Use case resumes at step 3
Exceptions	If the user, at any point, wishes to cancel, the user can type "cancel operation" and the use case resumes at step 1

Use Case: List Employees by Role – US15 (Specification)

Fig. 7. SSD: Center Administrator's UC 5

SSD: UC 5 – List Employees by Role

5.2 Actor: Receptionist

Fig. 8. Receptionist's Use Case Diagram

Table 7. Use Case: Register SNS User (US20)

Actor	Receptionist
Use case name	Register SNS User
Description	The receptionist registers a new SNS user in the system with their personal details.
Precondition	The "Receptionist" user must be selected.
Post-condition	A new SNSUser is saved in the system with a unique phone number, e-mail address, citizen card number, and SNS user number.
Main flow	→ System displays an option menu → User selects "Register SNS User" → System displays an input space → User fills the input space with name, date of birth, sex (optional), phone number, e-mail, citizen card number, and SNS user number → System checks if the information type is correct and if all unique fields are not already registered → Use case returns to step 3 until all mandatory properties are filled → A new SNSUser is added to the system
Alternative path	– Information type is incorrect or phone/CC not in Portuguese format → User receives an error message → Use case resumes at step 3 – Phone number, e-mail, CC number, or SNS number already exists → User receives a "Duplicate Data" error message → Use case resumes at step 3
Exceptions	If the user, at any point, wishes to cancel, the user can type "cancel operation" and the use case resumes at step 1

Use Case: Register SNS User – US20 (Specification)

Fig. 9. SSD: Receptionist's UC 1

SSD: UC 1 – Register SNS User

Table 8. Use Case: Schedule Vaccine Administration (US21)

Actor	Receptionist
Use case name	Schedule Vaccine Administration
Description	The receptionist schedules a vaccine administration appointment for a registered SNS user (in-person or phone scheduling).
Precondition	The "Receptionist" user must be selected, the SNS user must be registered, and at least one vaccine type must exist in the system.
Post-condition	A new Appointment is saved and linked to the SNS user and the selected vaccine type.
Main flow	→ System displays an option menu → User selects "Schedule Vaccine Administration" → System displays an input space → User inserts the SNS user's number → System checks if the SNS user is registered → System displays the available vaccine types → User selects a vaccine type and a date → System checks if the date is valid and the center has capacity → A new Appointment is added to the system
Alternative path	– SNS user number is not registered → User receives an error message → Use case resumes at step 3 – Selected date has no available capacity → System sends a "No Availability" message → Use case resumes at step 6
Exceptions	If the user, at any point, wishes to cancel, the user can type "cancel operation" and the use case resumes at step 1

Use Case: Schedule Vaccine Administration – US21 (Specification)

Fig. 10. SSD: Receptionist's UC 2

SSD: UC 2 – Schedule Vaccine Administration

Table 9. Use Case: Register SNS User Arrival (US22)

Actor	Receptionist
Use case name	Register SNS User Arrival
Description	The receptionist registers the arrival of an SNS user at the vaccination center to take the scheduled vaccine.
Precondition	The "Receptionist" user must be selected, the SNS user must be registered, and must have a scheduled appointment for the current day.
Post-condition	The SNS user is logically assigned to the waiting room.
Main flow	→ System displays an option menu → User selects "Register SNS User Arrival" → System displays an input space → User inserts the SNS user's number → System checks if the SNS user has a scheduled appointment for today → System checks that no duplicate arrival exists for the same user on the same day → The SNS user is added to the waiting room
Alternative path	– SNS user has no appointment scheduled for today → User receives an error message → Use case resumes at step 3 – SNS user arrival already registered for today → User receives a "Duplicate Arrival" error message → Use case resumes at step 3
Exceptions	If the user, at any point, wishes to cancel, the user can type "cancel operation" and the use case resumes at step 1

Use Case: Register SNS User Arrival – US22 (Specification)

Fig. 11. SSD: Receptionist's UC 3

SSD: UC 3 – Register SNS User Arrival

5.3 Actor: Nurse

Fig. 12. Nurse's Use Case Diagram

Table 10. Use Case: Consult Waiting Room (US40)

Actor	Nurse
Use case name	Consult Waiting Room
Description	The nurse consults the list of SNS users currently in the waiting room, ordered on a first-come, first-served basis.
Precondition	The "Nurse" user must be selected and at least one SNS user must be registered in the waiting room.
Post-condition	None
Main flow	→ System displays an option menu → User selects "Consult Waiting Room" → System checks if any SNS users are in the waiting room → System displays a list of SNS users in the waiting room in order of arrival
Alternative path	– No SNS users are in the waiting room → System sends a "Waiting Room Empty" message → Use case resumes at step 1
Exceptions	None

Use Case: Consult Waiting Room – US40 (Specification)

Fig. 13. SSD: Nurse's UC 1

SSD: UC 1 – Consult Waiting Room

Table 11. Use Case: Record Vaccine Administration (US41)

Actor	Nurse
Use case name	Record Vaccine Administration
Description	The nurse logs the administration of a vaccine to an SNS user, removes the specific dose from the inventory, and assigns the user to the recovery room.
Precondition	The "Nurse" user must be selected, at least one SNS user must be in the waiting room, and at least one matching vaccine dose must be available in inventory.
Post-condition	A VaccinationProcess is recorded, the administered dose is removed from available stock, and the SNS user is assigned to the recovery room.
Main flow	→ System displays an option menu → User selects "Record Vaccine Administration" → System displays the waiting room list → Nurse selects the SNS user to vaccinate → System displays the available vaccine doses matching the user's appointment → Nurse selects the specific dose by lot number → System removes the selected dose from inventory → A VaccinationProcess record is created and the SNS user is moved to the recovery room
Alternative path	– No matching vaccine doses are available in inventory → System sends a "No Stock Available" message → Use case resumes at step 3
Exceptions	If the user, at any point, wishes to cancel, the user can type "cancel operation" and the use case resumes at step 1

Use Case: Record Vaccine Administration – US41 (Specification)

Fig. 14. SSD: Nurse's UC 2

SSD: UC 2 – Record Vaccine Administration

5.4 Actor: SNS User

Fig. 15. SNS User's Use Case Diagram

Table 12. Use Case: Self-Schedule Administration (US30)

Actor	SNS User
Use case name	Self-Schedule Administration
Description	An SNS user uses the application directly to schedule a vaccine administration for a specific date.
Precondition	The "SNS User" profile must be selected and the user must be registered in the system.
Post-condition	A new Appointment is saved and linked to the SNS user and the selected vaccine type.
Main flow	→ System displays an option menu → User selects "Schedule Administration" → System displays an input space → User inserts their SNS user number → System checks if the SNS user is registered → System displays the available vaccine types → User selects a vaccine type and a desired date → System checks if the date is valid and the center has capacity → A new Appointment is created and confirmed to the user
Alternative path	– SNS user number is not registered → User receives an error message → Use case resumes at step 3 – Selected date has no available capacity → System sends a "No Availability" message → Use case resumes at step 6
Exceptions	If the user, at any point, wishes to cancel, the user can type "cancel operation" and the use case resumes at step 1

Use Case: Self-Schedule Administration – US30 (Specification)

Fig. 16. SSD: SNS User's UC 1

SSD: UC 1 – Self-Schedule Administration